Year 4 Project
[CSC1097]

Tag Battle Royale

[User Manual]

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1. Introduction

Tag Battle Royale is a mobile android app to facilitate games of tag.

It includes four main areas of usage:

- The Map Editor
- The Online Mode
- The Offline Mode
- The Replay Viewer

2. Installation guide

2.1 Android APK file

Requirements:

- Git installed
- Flutter 3.41.5
- Dart 3.11.3
- OpenJDK 21.0.10
- Android SDK 35.0.0
- ADB 36.0.2 (optional)
- An Android device

Instructions:

Step 1: Pull the repo

- git clone https://gitlab.computing.dcu.ie/slanksl2/2026-csc1097-tag-battle-royale.git

Step 2: Navigate to the android application directory

- cd 2026-csc1097-tag-battle-royale/src/tag_battle_royale/

Step 3: Install any dependencies

- flutter pub get

Step 4: Google Maps Api key

- https://developers.google.com/maps/flutter-package/config
- Put “MAPS_API_KEY=<YourAPIKey>” inside android/local.properties file

Step 5: Build the APK file

- flutter build apk --debug

Step 6: Install the APK file on your device

- Connect the device via wireless or usb debugging and run “flutter install”, if adb installed.
- Alternatively, upload the apk from “build/app/outputs/flutter-apk/app-debug.apk” to your device and install it from there.

2.2 Online Server

Tag Battle Royale online game server can be hosted locally within a Docker container. During testing Tag Battle Royale was hosted on a public server hosting site Fly.io. The necessary files for Fly.io are included with the server files.

- **2.2.1 Local Docker Hosting**

  Requirements:
  - Docker Desktop 4.0+ (Windows/Mac) or Docker Engine 20.10+ (Linux)
  - Port 3000 available

  Instructions:

  Step 1: Pull the repo
  - git clone https://gitlab.computing.dcu.ie/slanksl2/2026-csc1097-tag-battle-royale.git

  Step 2: Navigate to the game server directory
  - cd 2026-csc1097-tag-battle-royale/src/game-server/

  Step 3: Build the game server image
  - docker build -t tag-battle-royale .

  Step 4: Run the docker container
  - docker run -p 3000:3000 tag-battle-royale

  Step 5: Find your machine's local IP
  - The game server should be running on your machine on your local network using port 3000. You must find your machine's local IP address in order to connect.
  - ipconfig (Windows) or ip addr (Mac/Linux)
  - The full URL should typically look something like this: http://192.168.1.x:3000

  Step 6: Connect to the IP in app
  - A secret menu for configuring server URL is available in app: From the main menu, press the online button, and tap the cloud icon in the host/join menu five times. Input the server IP.
  - Alternatively, when building the app you can change the default server URL in /src/tag_battle_royale/main.dart. The constant is defined on line 32, and named defaultServerUrl.

- **2.2.2 Deploy To Fly.io**
  Requirements:
    - A Fly.io account (free tier is fine)
    - flyctl CLI

  Instructions:
  Step 1: Pull the repo
    - git clone https://gitlab.computing.dcu.ie/slanksl2/2026-csc1097-tag-battle-royale.git

  Step 2: Navigate to the game server directory
    - cd 2026-csc1097-tag-battle-royale/src/game-server/

  Step 3: Login to your Fly.io account
    - fly auth login

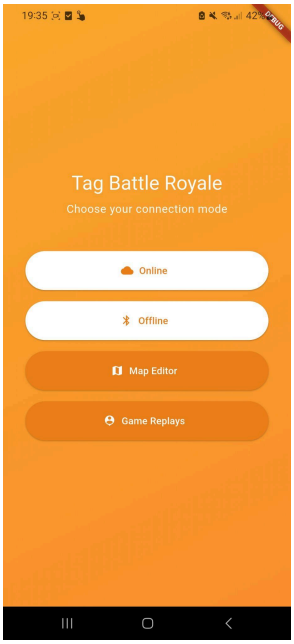
  Step 4: Initial setup to create the app on Fly.io's platform
    - fly launch
    - This command will ask you a few confirmation questions. We recommend for small-scale testing, no more than a shared-cpu-1x@256MB machine is required.

  Step 5: Deploy server
    - fly deploy
    - Fly.io will deploy two servers by default. We recommend for small-scale testing adding --ha=false parameter to the command to only deploy one.

  Step 6: Connect to the server in app
    - A secret menu for configuring server URL is available in app:
      From the main menu, press the online button, and tap the cloud icon in the host/join menu five times. Input the server URL given by Fly.io.
    - Alternatively, when building the app you can change the default server URL in /src/tag_battle_royale/main.dart. The constant is defined on line 32, and named defaultServerUrl.

Tag Battle Royale games can be run without an online server too, but strictly in offline mode.

3. Functionality



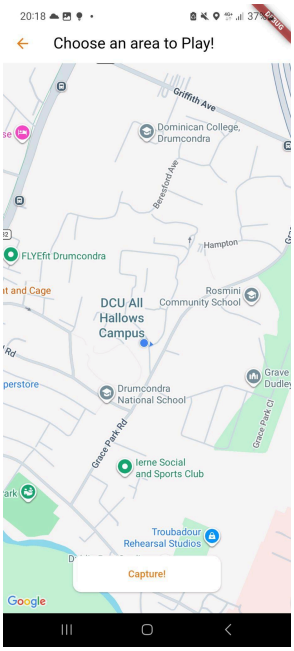
3.1 Home Page

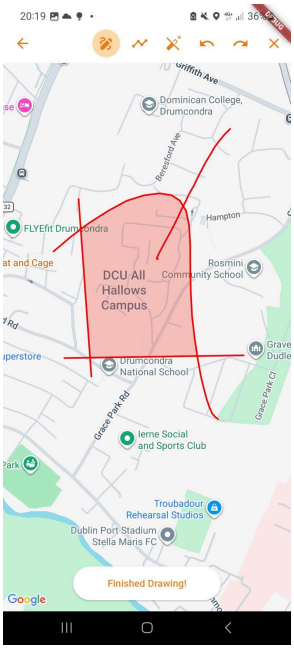
The Home page allows access to the four areas of Tag Battle Royale. Tag Battle Royale doesn't store any user data outside of the device so there is no signing up / signing in.

3.2 Map Editor

The Map Editor allows users to create new maps and view / delete already existing maps. Deleting a map will also delete any associated replays.When creating a new map the user is shown a map where they can select where to play. After pressing the capture button they can draw on the image to define the zones and name the map and zones. Finally they can save the map.

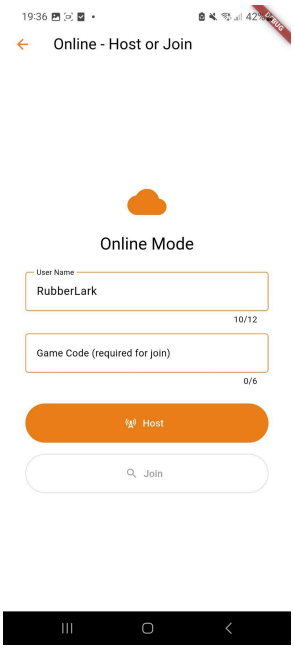


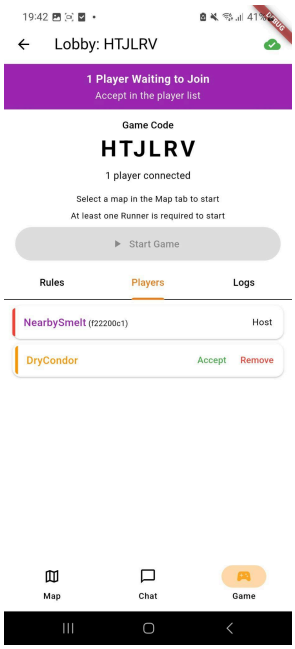




3.3 Online Mode

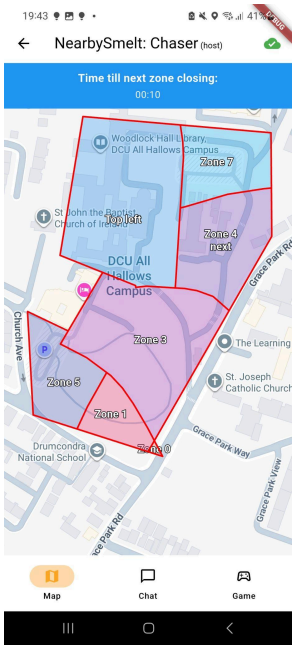
The Online Mode allows users to create or join a lobby. To join a lobby you must provide the lobby's code

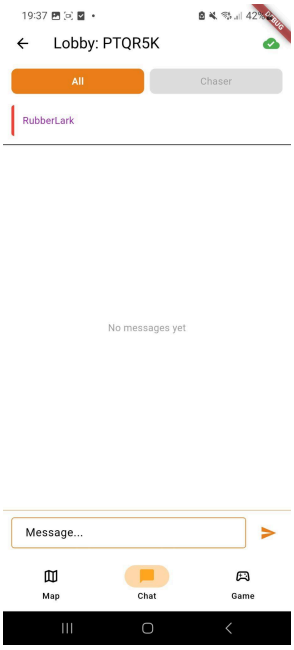


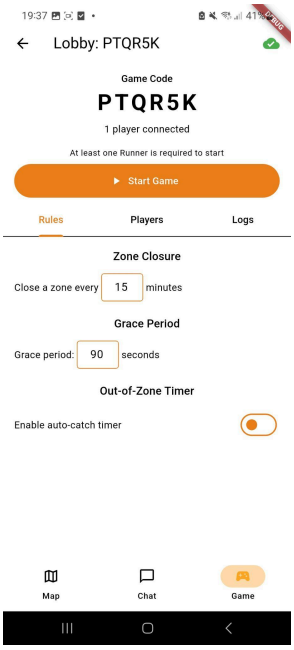


The lobby has three tabs; The Map tab, the Chat tab and the Game tab.
The Map tab allows the host to make a new map and/or pick a map for the game and order the zone closings. During the game the map will be displayed here along with the user and any other players visible to them (chasers can see other chasers and players not in an open zone). When playing, the host can change the zone order dynamically by pressing the zone they want to close next.

The Chat tab allows the players to communicate with each other. An all chat is provided where players can message each other but additionally, during the game runners have a dedicated runners chat and chasers have a chasers chat where only players of the corresponding role can access that chat.





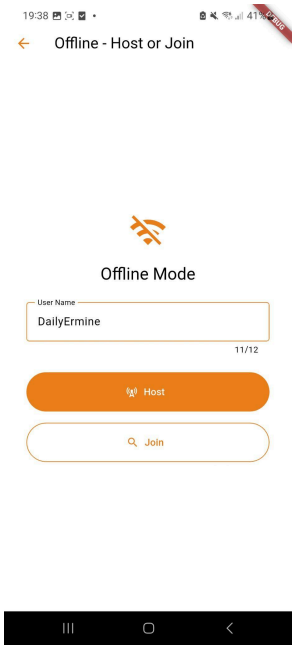


The game tab allows the host to accept incoming players, kick players, change the game settings and start the game.The Online lobby automatically combines the players positions and saves the replay and map files

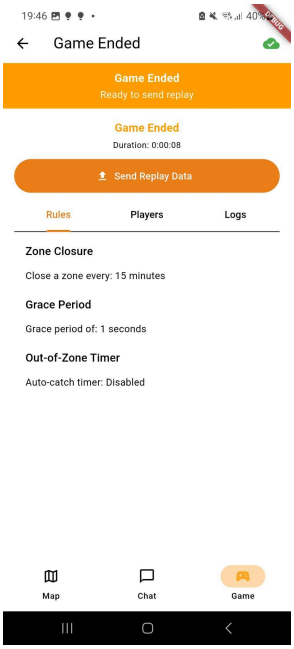
3.4 Offline Mode

The Offline Mode is similar to the Online Mode, instead of connecting to the backend server with a game code, the players connect directly to each other.

The Offline Mode lobby is the same as the Online Mode lobby before playing, excusing that the host cannot create a new map in the Offline Mode. This includes the Map tab, the Chat tab and the Game tab.







When playing in the Offline Mode, the players disconnect from each other. Other players are not displayed and the host cannot change the zone closure order dynamically. Each player must manually end the game on their device. Additionally, to make the replay, the players must reconnect to the host, where the replay is automatically saved along with the map.

3.5 Replay Viewer

The Replay Viewer allows users to view and delete replays saved to their device. The replays are created and saved when playing the game in an Offline or Online game.

When viewing a replay the user is shown controls to play / pause the replay, display only certain players and toggle players nameplates. The user can also minimise the controls



